

———— // ZPWR-MIDI-FX // ————

zpwr-midi-fx

v0.2.8

MODULE REFERENCE

wire your own MIDI processor from note-stream blocks

2026-06-23

MenkeTechnologies

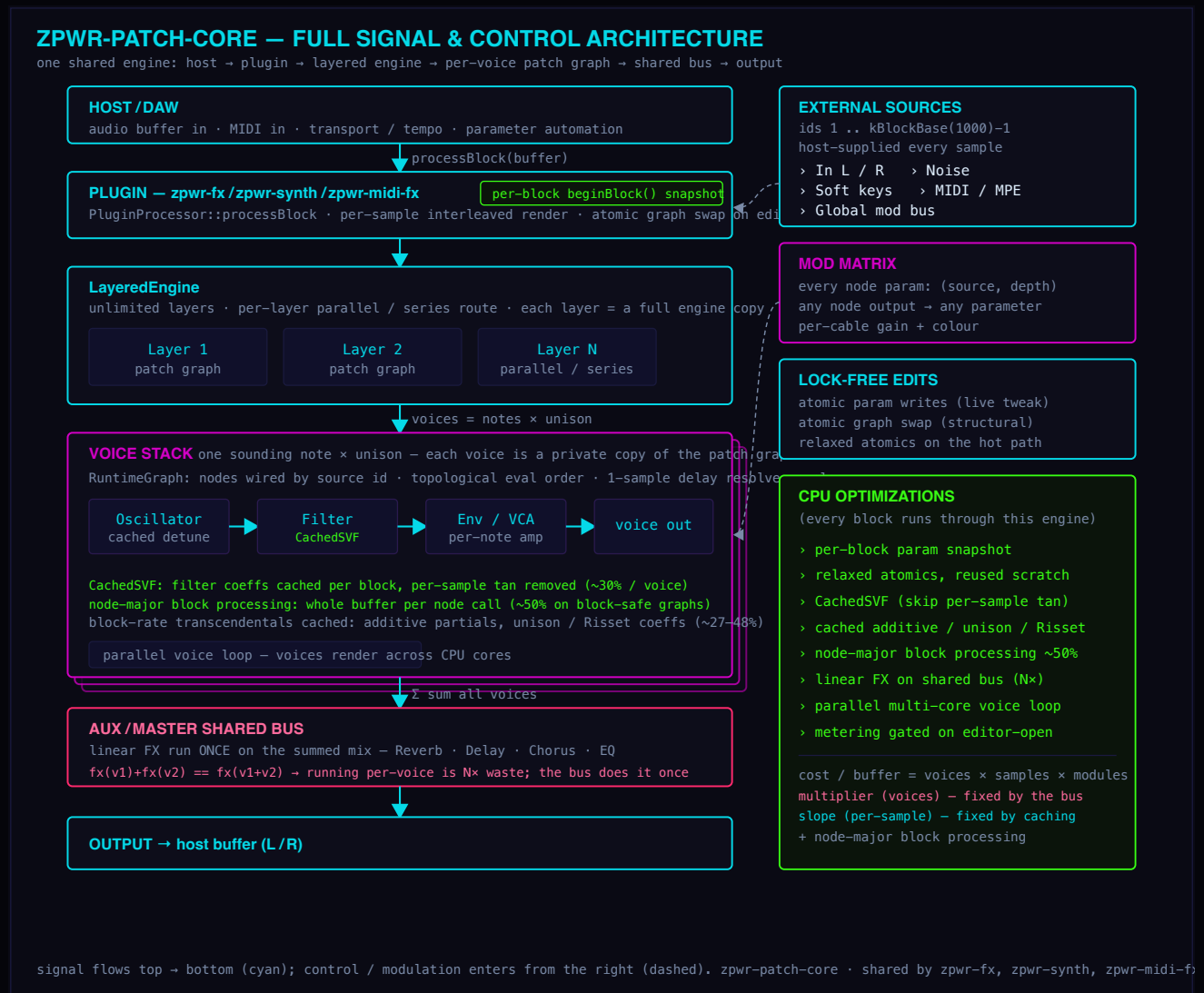
———— MENKETECHNOLOGIES ————

Contents

Architecture	2
Signal Sources	3
Modulation	4
Dynamics	6
Control	8
Expression	9
Harmony	10
Probability	14
Routing	15
Sequencing	17
Tuning	25
Voicing	26

Architecture

ZPWR-MIDI-FX runs on the shared zpwr-patch-core engine. The full signal and control path – from the host buffer, through the layered engine and the per-voice patch graph, onto the shared bus, and back out – is shown below.



Signal Sources

External sources you can cable into any block input or modulation route.

1 MIDI In 3 Random Mod Wheel Pitch Bend Aftertouch Velocity Expression Sustain MPE Bend MPE Pressure MPE Slide

Modulation

7 modules

Generative 3 inputs

Probabilistic melodic random walk synced to a rate.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	—	—
Range	1 - 12	3	st
Low	0 - 127	48	—
High	0 - 127	72	—
Probability	0 - 1	0.7	—

MidiLFO 3 inputs

Low-frequency modulation source (sine/tri/saw/square) on the scalar control output.

Param	Range	Default	Unit
Rate	0.01 - 20	1	Hz
Shape	Sine · Triangle · Saw · Square	—	—
Depth	0 - 1	1	—

MidiEnv 3 inputs

ADSR envelope triggered by the note gate, on the scalar control output.

Param	Range	Default	Unit
Attack	1 - 5000	5	ms
Decay	1 - 5000	200	ms
Sustain	0 - 1	0.7	—
Release	1 - 5000	300	ms

MidiRandom 3 inputs

Generates random notes within a pitch range at a synced rate.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	—	—
Low Note	0 - 127	48	—
High Note	0 - 127	72	—
Gate	0 - 2	0.5	—
Velocity	1 - 127	100	—

MidiSampleHold 3 inputs

Samples and holds the incoming control value on each note.

MidiSlew 3 inputs

Smooths (glides) the control value over a set time.

Param	Range	Default	Unit
Time	0 - 2000	100	ms

RandOctave 3 inputs

Randomly shifts notes by up to +/- Range octaves with a chance.

Param	Range	Default	Unit
Range	1 - 3	1	-
Chance	0 - 1	1	-

Dynamics

8 modules

Velocity 3 inputs

Scales and offsets note velocity, with a modulation amount.

Param	Range	Default	Unit
Scale	0 - 2	1	-
Offset	-64 - 64	0	-
Mod Amt	-64 - 64	0	-

VelCurve 3 inputs

Reshapes velocity through a power curve.

Param	Range	Default	Unit
Curve	0.2 - 5	1	-

Humanize 3 inputs

Adds random timing and velocity jitter for a human feel.

Param	Range	Default	Unit
Timing	0 - 50	8	ms
Velocity	0 - 1	0.3	-

Accent 3 inputs

Boosts velocity on every Nth note.

Param	Range	Default	Unit
Every	1 - 16	4	-
Amount	0 - 1	0.5	-

VelClip 3 inputs

Clamps velocity between Min and Max.

Param	Range	Default	Unit
Min	1 - 127	1	-
Max	1 - 127	127	-

Ramp 3 inputs

Velocity ramp over a run of notes, ascending or descending.

Param	Range	Default	Unit
Length	2 - 16	8	-
Depth	0 - 1	0.6	-
Direction	Up · Down		-

VelCompress 3 inputs

Velocity compressor: velocities above Threshold are pulled toward it by Ratio (1 = off, 8 = hard), narrowing the dynamic range smoothly instead of hard-clamping like VelClip.

Param	Range	Default	Unit
Threshold	1 - 127	64	-
Ratio	1 - 8	2	-

VelInvert 3 inputs

Flips the velocity scale around its midpoint (soft <-> loud) by Amount, so your accents become ghost notes and the ghosts become accents - a velocity mirror, blendable to taste.

Param	Range	Default	Unit
Amount	0 - 1	1	-

Control

5 modules

CCMap 3 inputs

Remaps a CC number and scales/offsets its value.

Param	Range	Default	Unit
From	-1 - 127	1	-
To	0 - 127	11	-
Scale	0 - 2	1	-
Offset	-1 - 1	0	-

Note2CC 3 inputs

Emits a CC from each note (pitch or velocity).

Param	Range	Default	Unit
CC	0 - 127	1	-
Source	Pitch · Velocity	-	-

CC2Note 3 inputs

A watched CC crossing a threshold triggers a note.

Param	Range	Default	Unit
Watch	0 - 127	1	-
Note	0 - 127	60	-
Threshold	0 - 1	0.5	-

BendProc 3 inputs

Scales and/or inverts incoming pitch-bend.

Param	Range	Default	Unit
Scale	0 - 2	1	-
Invert	0 - 1	0	-

ATProc 3 inputs

Scales aftertouch, optionally converting it to a CC.

Param	Range	Default	Unit
Scale	0 - 2	1	-
To CC	-1 - 127	-1	-

Expression

3 modules

Scoop 3 inputs

Bends up into each struck note from Depth below, resolving to pitch over Time - the vocal/brass scoop, a pitch-bend ramp that ends at centre so no bend is left hanging.

Param	Range	Default	Unit
Depth	0 - 1	0.3	-
Time	1 - 1000	80	ms

Fall 3 inputs

Jazz fall-off: on release the pitch sweeps down by Depth over Time while the note rings, then the note releases and the bend snaps back to centre. The note-off is delayed to the end of the fall.

Param	Range	Default	Unit
Depth	0 - 1	0.5	-
Time	10 - 2000	300	ms

MidiVibrato 3 inputs

A pitch-bend LFO that fades in after an Onset delay while any note is held - the delayed, played-in vibrato of a string or voice rather than an instant wobble. Rate in Hz, Depth as bend amount.

Param	Range	Default	Unit
Rate	0.1 - 12	5	Hz
Depth	0 - 1	0.15	-
Onset	0 - 2000	300	ms

Harmony

19 modules

Chord 3 inputs

Turns each incoming note into a chord (165 types) with inversion, spread, octave doubling and strum.

Param	Range	Default
Type	Unison · Octave · 5th (Power) · 5th + Oct · Tritone · Major · Minor · Diminished · Augmented · Major (1st inv) · Major (2nd inv)	
Inversion	0 - 4	0
Spread	0 - 2	0
Octave x2	0 - 2	0
Transpose	-24 - 24	0
Strum	0 - 200	0

Scale 3 inputs

Quantizes incoming notes to the nearest degree of the chosen root and scale.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentatonic	

Transpose 3 inputs

Shifts every note by a fixed semitone amount plus a modulatable offset.

Param	Range	Default	Unit
Semitones	-24 - 24	0	st
Mod Amt	-24 - 24	0	st

Harmonize 3 inputs

Adds up to three fixed-interval harmony voices to each note.

Param	Range	Default	Unit
Interval 1	-24 - 24	7	st
Interval 2	-24 - 24	0	st
Interval 3	-24 - 24	0	st

Drone 3 inputs

Doubles every note with a fixed pedal root.

Param	Range	Default	Unit
Root	0 - 127	36	-

MidiCluster 3 inputs

Adds symmetric tone-cluster voices above and below each note.

Param	Range	Default	Unit
Width	0 - 4	1	—
Spread	1 - 4	1	st

Glissando 3 inputs

Fills a stepwise run between consecutive notes.

Param	Range	Default	Unit
Rate	5 - 200	40	ms

ChordMem 3 inputs

Each key plays a stored chord shape (semitone intervals from node config).

ChordProg 3 inputs

Each note triggers the next chord in a stored root progression (node config).

ChordVoicing 3 inputs

Re-voices each note into a jazz shape (open / shell / quartal / spread).

Param	Range	Default	Unit
Voicing	Open · Shell · Quartal · Spread	—	—

Octave 3 inputs

Doubles each note up and/or down by whole octaves.

Param	Range	Default	Unit
Up	0 - 3	1	—
Down	0 - 3	0	—

MidiFold 3 inputs

Folds out-of-range notes back inside a Low..High window.

Param	Range	Default	Unit
Low Note	0 - 127	48	—
High Note	0 - 127	72	—

Invert 3 inputs

Mirrors pitch around a pivot note.

Param	Range	Default	Unit
Pivot	0 - 127	60	-

Diatonic 3 inputs

Transposes by scale DEGREES within Root/Scale (a third, a fifth...) instead of by raw semitones like Transpose - the shift always lands back in key. Mod Amt adds a modulatable degree offset.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · P	
Degrees	-7 - 7	2
Mod Amt	-7 - 7	0

DiatonicChord 3 inputs

Auto-harmoniser: builds the in-key chord on every note by stacking diatonic thirds (triad, 7th...) along Root/Scale, so single-finger melodies become correct chords in the key.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pen	
Notes	2 - 4	3

NeoRiemann 3 inputs

Neo-Riemannian triad transforms: builds a major/minor triad on each note and applies P (parallel), L (leading-tone) or R (relative) - the maximally-smooth chord moves of late-Romantic and film harmony, each sharing two common tones with the input.

Param	Range	Default	Unit
Quality	Major · Minor		-
Transform	None · P · L · R		-

Counterpoint 3 inputs

Adds a second voice in contrary motion - when the melody rises the counter voice falls and vice-versa - quantised to Root/Scale. Interval sets the starting separation.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian ·	
Interval	1 - 24	7

Bassline 3 inputs

Follows the chord with a mono bass voice: the lowest held note dropped Octaves down, retriggered whenever the lowest note changes. Passes the original notes through and adds the bass beneath them.

Param	Range	Default	Unit
Octaves	1 - 3	1	-

Param	Range	Default	Unit
Velocity	1 - 127	100	-

VoiceLead 3 inputs

Smooth voice leading: places each note's pitch class in the octave nearest the previous output, so a line or chord progression moves by the smallest interval. Range caps the leap; past it the original octave is kept.

Param	Range	Default	Unit
Range	1 - 12	7	st

Probability

1 modules

Chance 3 inputs

Probabilistic gate: lets each note through with the set probability.

Param	Range	Default	Unit
Probability	0 - 1	0.8	-
Mod Amt	-1 - 1	0	-

Routing

9 modules

Split 3 inputs

Keyboard split: notes below the split point transpose by Low, the rest by High.

Param	Range	Default	Unit
Split	0 - 127	60	-
Low	-24 - 24	0	st
High	-24 - 24	0	st

Transformer 3 inputs

Rule: notes within a range are transposed and velocity-scaled.

Param	Range	Default	Unit
Low	0 - 127	0	-
High	0 - 127	127	-
Transpose	-24 - 24	0	st
VelScale	0 - 2	1	-

Merge 3 inputs

Passes both note-stream inputs through, merged into one stream.

NoteFilter 3 inputs

Passes only notes inside a key and velocity window.

Param	Range	Default	Unit
Low Note	0 - 127	0	-
High Note	0 - 127	127	-
Low Vel	1 - 127	1	-
High Vel	1 - 127	127	-

Midilatch 3 inputs

Holds notes after release until the next note arrives.

Channel 3 inputs

Reassigns notes to a fixed MIDI channel.

Param	Range	Default	Unit
Channel	1 - 16	1	-

FixedNote 3 inputs

Forces every note to a single fixed pitch (drum / trigger).

Param	Range	Default	Unit
Note	0 - 127	36	-

KeySwitch 3 inputs

Splits the keyboard at a note; one zone passes, the other is filtered.

Param	Range	Default	Unit
Split	0 - 127	60	-
Low Zone	High passes · Low passes		-

NoteLimit 3 inputs

Polyphony cap: never lets more than Max notes sound at once, stealing the oldest (FIFO note-off) when a new note would exceed the limit - tames dense chords and runaway generators.

Param	Range	Default	Unit
Max	1 - 16	4	-

Sequencing

35 modules

Arp 3 inputs

Arpeggiator: cycles held notes through a pattern at a synced rate with gate, octaves, swing and step count. Latch holds the chord so it keeps arpeggiating after the keys are released (the next key starts a new chord).

Param	Range	Default	Unit
Mode	Up · Down · Up/Down · Down/Up · Converge · Diverge · As Played · Random · Chord		–
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0 – 2	0.5	–
Octaves	1 – 4	1	–
Swing	0 – 0.75	0	–
Steps	1 – 16	16	–
Latch	0 – 1	0	–

SeqEuclid 3 inputs

Euclidean rhythm gate: spreads Pulses evenly across Steps (rotatable) at a synced rate.

Param	Range	Default	Unit
Pulses	1 – 16	5	–
Steps	1 – 16	8	–
Rotate	0 – 15	0	–
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0 – 2	0.5	–

MidiGameOfLife 3 inputs

Conway's Game of Life MIDI sequencer: an 8x8 cell grid evolves under the B3/S23 rule; each clock step reads the next column and plays a note for every live cell (row maps to a scale degree from Root/Scale), and a full 8-step sweep advances one generation. Gliders, blinkers and still-lives become ever-shifting melodic and harmonic patterns. Density sets the random fill when the grid (re)seeds. The MIDI-note counterpart to the GameOfLife CV block.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · P	
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
Gate	0 – 2	0.5
Density	0 – 1	0.35

MidiBrianBrain 3 inputs

Brian's Brain MIDI sequencer: an 8x8 three-state grid (ready / firing / dying) evolves under Brian's Brain rules - a ready cell ignites only with exactly two firing neighbours, firing cells pass to dying, dying cells reset. Each clock step plays a note for every firing cell in the next column (row maps to a scale degree from Root/Scale); a full 8-step sweep advances one generation. The refractory state keeps it restless and sparkly where Game of Life settles. Density sets the random fill. The MIDI-note counterpart to the BrianBrain CV block.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · P	
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
Gate	0 - 2	0.5
Density	0 - 1	0.3

Midilangton 3 inputs

Langton's Ant MIDI melody: a single turmite walks an 8x8 grid - turning right on a white cell, left on a black one, flipping each cell as it leaves, then stepping forward. Each clock advances the ant Steps cells and plays one note for its current row (mapped to a scale degree from Root/Scale), so the melody scribbles chaotically then locks into the famous periodic 'highway'. Emergent order from one trivial rule - a single moving agent, not a population.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pen	
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
Gate	0 - 2	0.5
Steps	1 - 16	1

Swing 3 inputs

Pushes off-beat eighth-notes later for a swing/shuffle feel.

Param	Range	Default	Unit
Amount	0 - 0.75	0.3	-

PatternGate 3 inputs

Gates note-ons through a 16-step on/off pattern (bitmask).

Param	Range	Default	Unit
Pattern	0 - 65535	21845	-
Steps	1 - 16	16	-

NoteOffDelay 3 inputs

Extends each note's gate by delaying its note-off.

Param	Range	Default	Unit
Delay	0 - 2000	200	ms

Ricochet 3 inputs

Accelerating bouncing retriggers with decaying velocity.

Param	Range	Default	Unit
Bounces	1 - 16	6	-
Decay	0.3 - 0.98	0.7	-
Start	10 - 500	100	ms

StepSeqMidi 3 inputs

Melodic step sequencer - semitone offsets (node config) clock the held root.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
Gate	0.05 - 1	0.5	—
Velocity	1 - 127	100	—

Echo 3 inputs

MIDI delay: repeats notes at a synced time with velocity feedback decay.

Param	Range	Default	Unit
Time	1 - 2000	250	ms
Feedback	0 - 0.95	0.5	—
Repeats	1 - 8	3	—

Flam 3 inputs

Precedes every note with a quiet grace note, delaying the main hit so the grace leads it - the classic drum flam.

Param	Range	Default	Unit
Time	1 - 200	30	ms
Grace Vel	0 - 1	0.5	—

Roll 3 inputs

Drum roll / buzz: retriggers every held note at Rate for as long as it is held, each hit gated to a fraction of the interval.

Param	Range	Default	Unit
Rate	1 - 50	12	Hz
Gate	0.05 - 0.95	0.5	—

Trill 3 inputs

Keyboard trill: while a note is held, alternates rapidly between it and the note Interval semitones above, at Rate.

Param	Range	Default	Unit
Rate	1 - 50	10	Hz
Interval	1 - 12	2	st
Gate	0.05 - 0.95	0.5	—

Mordent 3 inputs

One-shot ornament: each struck note plays principal -> auxiliary -> principal at the attack, then sustains, the single-flick counterpart to the continuous Trill.

Param	Range	Default	Unit
Time	5 - 150	40	ms
Interval	1 - 12	2	st

Param	Range	Default	Unit
Direction	Upper · Lower		–

Turn 3 inputs

Gruppetto turn: a four-step ornament at the attack - upper auxiliary, principal, lower auxiliary, then the principal held. Time sets each step, Interval the neighbour distance.

Param	Range	Default	Unit
Time	5 - 150	35	ms
Interval	1 - 12	2	st

Appoggiatura 3 inputs

Leaning grace note: each struck note first sounds a long auxiliary (Interval away) that takes Time from the principal, which then sustains - the expressive long grace, unlike the quick Flam or Mordent.

Param	Range	Default	Unit
Time	20 - 400	120	ms
Interval	1 - 12	2	st
Direction	Upper · Lower		–

Strum 3 inputs

Spreads a chord's notes over time, up or down, like a guitar strum.

Param	Range	Default	Unit
Spread	0 - 200	20	ms
Direction	Up · Down		–

MidiQuantize 3 inputs

Snaps note starts to a synced grid by a strength amount.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Strength	0 - 1	1	–

SeqRatchet 3 inputs

Retriggers each note Count times within a synced step.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Count	1 - 8	4	–
Gate	0.05 - 1	0.9	–

NoteLength 3 inputs

Forces a fixed note duration (gate length).

Param	Range	Default	Unit
Length	1 - 2000	200	ms

Cascade 3 inputs

Each struck note fans out into a timed run of Steps notes climbing (or falling) the scale from it - an arpeggiated flourish fired per key, unlike Arp which cycles the whole held chord.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian ·	
Steps	2 - 8	4
Time	5 - 200	50
Direction	Up · Down	

MidiTuring 3 inputs

Turing-machine sequencer (Music Thing style): a circular shift register clocks once per step and the held notes fire when the read bit is 1. Change is the probability the recycled bit flips - 0 locks an evolving loop, 1 fully randomises, between = slowly mutating melodies.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		-
Length	2 - 16	8	-
Change	0 - 1	0.15	-
Gate	0.05 - 1	0.5	-

Polymeter 3 inputs

Two step lanes of different lengths (LenA, LenB) clock together over the held notes, phasing against each other so the combined pattern only repeats after lcm(LenA,LenB) steps - instant polymetric movement from one chord.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		-
Len A	1 - 16	3	-
Len B	1 - 16	4	-
Gate	0.05 - 1	0.5	-

MidiTranceGate 3 inputs

Chops the held notes with a 16-step on/off Pattern at a synced Rate: every 'on' step retriggers all held notes, gated to a fraction of the step. The rhythmic gate behind the trance sound - pattern-driven, where Roll is a fixed pulse.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		-
Pattern	0 - 65535	21845	-
Steps	1 - 16	16	-
Gate	0.05 - 1	0.5	-

StepArp 3 inputs

Cthulhu-style step arpeggiator: walks the held chord strictly upward across Octaves octaves at a synced Rate, accenting the first step of each cycle - the rolling, octave-jumping arp of Xfer Cthulhu's sequencer. Distinct from Arp's mode-cycling: this climbs the chord in order.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0.05 – 1	0.5	–
Octaves	1 – 4	2	–
Accent	1 – 127	110	–

ChordArp 3 inputs

Cthulhu-style chord arpeggiator: each key builds a full chord (Type) and the result is arpeggiated upward across Octaves at a synced Rate - one finger plays a moving arpeggio of the whole chord. Releasing the key stops its run.

Param	Range	Default
Type	Unison · Octave · 5th (Power) · 5th + Oct · Tritone · Major · Minor · Diminished · Augmented · Major (1st inv) · Major (2nd inv)	
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
Gate	0.05 – 1	0.5
Octaves	1 – 4	1

OctaveArp 3 inputs

Cthulhu's arp octave lane: arpeggiates the held chord while a per-step octave-offset Pattern leaps the line up and down by whole octaves (alternating, staircase, bounce...) instead of climbing monotonically like StepArp.

Param	Range	Default	Unit
Pattern	0 – 7	1	–
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0.05 – 1	0.5	–

GateArp 3 inputs

Cthulhu's arp gate lane: walks the held chord upward across Octaves while the note length cycles a per-step staccato/legato pattern, so the rhythm breathes - short stabs then sustained notes. Gate scales the pattern (>1 overlaps into the next step).

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0.25 – 2	1	–
Octaves	1 – 4	1	–

VelArp 3 inputs

Cthulhu's arp velocity lane as a ramp: arpeggiates the held chord while velocity sweeps from From to To across Steps then resets, so every pass swells or fades without a per-step editor.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0.05 – 1	0.5	–
From	1 – 127	40	–

Param	Range	Default	Unit
To	1 - 127	120	-
Steps	1 - 32	8	-

TransArp 3 inputs

Cthulhu's arp transpose lane: arpeggiates the held chord upward across Octaves while a per-step semitone Pattern (fifth pedals, triad arps, scalar runs) is added on top, so the line outlines a melodic contour, not just the chord tones.

Param	Range	Default	Unit
Pattern	0 - 7	2	-
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Gate	0.05 - 1	0.5	-
Octaves	1 - 4	1	-

RatchetArp 3 inputs

Cthulhu's arp repeat lane: walks the held chord upward across Octaves while a per-step pattern retriggers selected steps multiple times (capped by Max), packing rolls into the line. Unlike SeqRatchet's fixed count, the repeat count varies per step.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Gate	0.05 - 1	0.5	-
Octaves	1 - 4	1	-
Max	1 - 8	4	-

SkipArp 3 inputs

Cthulhu's arp skip lane: arpeggiates the held chord upward across Octaves, but each step has a Skip probability of being dropped to a rest, thinning the pattern differently every cycle for generative movement.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Gate	0.05 - 1	0.5	-
Octaves	1 - 4	1	-
Skip	0 - 0.95	0.3	-

Drunk 3 inputs

Random-walk melodic sequencer: each synced Rate step nudges the pitch by a random +/-Step semitones, bounded within +/-Range of the lowest held note - Brownian melody anchored to the chord, unlike MidiRandom's memoryless picks.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Gate	0.05 - 1	0.5	-
Step	1 - 7	2	st
Range	1 - 24	12	st

EuclidMel 3 inputs

Euclidean melody: spreads Pulses evenly across Steps (Bjorklund) and, on each hit, advances a cursor through the held chord - the rhythm is Euclidean and the pitch walks the chord, unlike SeqEuclid which stabs all held notes per pulse.

Param	Range	Default	Unit
Pulses	1 - 16	5	-
Steps	1 - 16	8	-
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		-
Gate	0.05 - 1	0.5	-

Tuning

1 modules

Microtuner 3 inputs

Stretched-octave microtuning via per-note pitch detune.

Param	Range	Default	Unit
Stretch	-50 - 50	0	cents/oct

Voicing

5 modules

MPEConvert 3 inputs

Assigns each note its own rotating channel for per-note MPE expression.

Param	Range	Default	Unit
Voices	2 - 15	15	-

Sustain 3 inputs

Sustain-pedal simulation: while Hold is engaged, note-offs are captured so the notes keep sounding; when Hold drops, every held note releases at once. Hold is a parameter, so a CC, an LFO or automation can play the pedal.

Param	Range	Default	Unit
Hold	Off · On		-

Mono 3 inputs

Monophonic note priority (last / lowest / highest).

Param	Range	Default	Unit
Priority	Last · Lowest · Highest		-

MidiUnison 3 inputs

Doubles each note into multiple voices, optionally spread across channels.

Param	Range	Default	Unit
Voices	1 - 4	2	-
Spread	Same chan · Spread chans		-

Slide 3 inputs

Cthulhu's slide: mono legato glue. Picks one held note by Priority (last/low/high) and emits the new note-on before releasing the old one, so a portamento synth slurs between notes instead of retriggering.

Param	Range	Default	Unit
Priority	Last · Low · High		-